№5.20

Дано:

$\Delta P = 100$ атм

$\frac{\Delta V}{V} = 0{,}035\%$

$\beta_{T_T} = 3{,}9 \cdot 10^{-11}\ Па^{-1}$

Найти: $\frac{C_p}{C_v}$

Решение:

1) $\beta_{T_T} = \left(\frac{\partial V}{\partial p}\right)_T V^{-1}$; $\beta_{T_S} = \left(\frac{\partial V}{\partial p}\right)_S V^{-1}$

2) $C_p = \left(\frac{\partial Q}{\partial T}\right)_p = T\left(\frac{\partial S}{\partial T}\right)_p = T\left(\frac{\partial T}{\partial P}\right)_s$

3) $\left(\frac{\partial p}{\partial V}\right)_S = \gamma\left(\frac{\partial p}{\partial V}\right)_T$ $\quad \left(\frac{\partial V}{\partial T}\right)_p = \left(\frac{\partial V}{\partial S}\right)_p \left(\frac{\partial S}{\partial T}\right)_p$

$$\gamma = \frac{C_p}{C_v} = \frac{dp_s}{dV_s} \cdot \beta V = \frac{100 \cdot 10^5 \cdot 3{,}9 \cdot 10^{-11}}{0{,}00035} = 1{,}13$$

№ 5.42

Дано:

$F = -AVT^4$

$P = 1$ атм

$V = 1$ л

Найти: C_v

Решение:

1) $C_v = T\left(\frac{\partial S}{\partial T}\right)_V$ $\quad \left(\frac{\partial T}{\partial P}\right)_V = \left(\frac{\partial T}{\partial S}\right)_V \left(\frac{\partial S}{\partial P}\right)_V$

2) $F = U - TS \Rightarrow dF = -pdV - SdT$

$$S = -\left(\frac{\partial F}{\partial T}\right)_V = 4AVT^3$$

3) $C_v = T\left(\frac{\partial S}{\partial T}\right)_V = 12AVT^3$, $P = \frac{1}{3}aT^4 \Rightarrow T^4 = \frac{3p}{a}$

$\Rightarrow T = 141190$ K

$$C_v = 12AV \cdot \left(\frac{3p}{a}\right)^{\frac{3}{4}} = 8{,}5 \cdot 10^{-3}\ \frac{Дж}{К}$$

2) $C_{V\,\text{ид}} = C_V \cdot \nu = C_V \cdot \dfrac{pV}{RT} \approx 10^{-3}\ \dfrac{\text{Дж}}{\text{К}}$

$\Rightarrow C_V = 8{,}5\, C_{V\,\text{ид}}.$

N 5.20

Продолжение:

$$\left(\frac{\partial P}{\partial V}\right)_S = \left(\frac{\partial P}{\partial T}\right)_S \left(\frac{\partial T}{\partial V}\right)_S = \left(\frac{\partial P}{\partial S}\right)_T \cdot \underbrace{\left(\frac{\partial S}{\partial T}\right)_P}_{= C_P} \cdot \left(\frac{\partial S}{\partial V}\right)_T \cdot \underbrace{\left(\frac{\partial T}{\partial S}\right)_V}_{= \frac{1}{C_V}}$$

$$\left(\frac{\partial P}{\partial V}\right)_S = \frac{C_P}{C_V} \cdot \left(\frac{\partial P}{\partial V}\right)_T = \frac{C_P}{C_V} \cdot \frac{1}{\beta V}$$

$$\gamma = \frac{dP_S}{dV_S} \cdot V \cdot \beta = 1{,}13.$$

N 12.13

Дано:

$D = 0{,}2$ м

$d = 10^{-3}$ м

$\sigma = 70$ дин/см

Найти: m_{max}

Решение:

$$P = \frac{F}{S} = \frac{2\pi R \sigma}{\frac{\pi d^2}{4}} = \frac{mg}{S}$$

$\Rightarrow$

$$\frac{4\sigma}{d} = \frac{mg}{S} \Rightarrow m = \frac{4S\sigma}{dg} = \frac{\pi D^2 \sigma}{gd} \approx 902\ \text{г}$$

Масса не зависит от ж-ти.
